

GALLANT IAS ACADEMY
PRELIMS CUM MAINS BATCH 2027
GEOGRAPHY
CLIMATOLOGY
MODULE 3
DISTRIBUTION OF TEMPERATURE

INSOLATION

Insolation refers to the "**incoming solar radiation**" received by the Earth's surface. It is a combination of the words "incoming" and "solar radiation". Insolation is received as short wavelength (**shortwave**) by the Earth and **terrestrial radiation** emitted by the Earth is in the form of long wavelength (**longwave**). Insolation represents the **total amount of solar energy received per unit area, usually measured in watts per square meter (W/m^2)**.

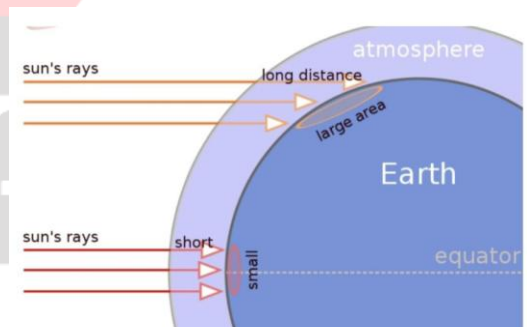
FACTORS AFFECTING THE DISTRIBUTION OF INSOLATION

a) Angle of Incidence of Sun's Rays

The **angle between the rays of the sun and the surface of the earth** at a given place largely **determines the amount of insolation** to be received at that place. The curved surface associated with the spherical shape of the earth presents a continually varied angle to the incoming parallel rays of the Sun. This latitudinal variation in the angle of solar rays results in an uneven global distribution of insolation.

The sun's rays are **more or less vertical (maximum angle of 90°) at the lower latitudes** and become **more and more oblique poleward**. As the angle of inclination decreases poleward, the amount of insolation received also decreases.

- **Oblique rays have to pass through thicker portion of the atmosphere** than the vertical rays. Thus, the oblique rays have to travel larger distances than the vertical rays. Consequently, the amount of solar energy lost due to reflection, scattering and absorption increases with increasing distance of travel path covered by the sun's rays through the atmosphere.
- **Vertical rays are spread over minimum area** of the earth's surface, and they heat the minimum possible area and thus the **energy received per unit area increases**. On the other hand, **oblique rays are spread over larger area of the earth's surface** and hence the amount of **energy received per unit area decreases**.



Other than latitude, the angle of incidence can vary according to seasons and time of the day. At **mid-day the intensity of insolation is maximum** but, in the morning and evening hours it is reduced because of the slanting rays of the sun (lower angle of incidence). Similarly, most of the **latitudes in winter season the amount of insolation at the surface is lesser** due to the lower angle of incidence.

b) Length of Day (Duration of Sunshine)

Longer period of sunshine ensures larger supply of radiation to a particular area. As the seasons change, the length of day varies at all places except at the equator. The latitude exercises the most dominant control over the length of the day. For example, **the higher latitudes can have longer durations of sunshine in the summer season.**

c) Distance Between the Earth and the Sun

The distance between the sun and the earth **changes during course of a year** because the earth revolves around the sun in an **elliptical orbit**. As per rule, the earth at the time of **perihelion, when it is nearest to the sun**, should receive **maximum insolation** while it should receive **minimum insolation** at the time of aphelion when the earth is at the greatest distance from the sun. Although the amount of insolation received by the boundary of the Earth's atmosphere is **slightly higher in January than in July (7%)**, there are **other major factors** such as angle of incidence and duration of sunshine which mostly control the distribution of insolation received in an area.

d) Sunspots (Variations in Solar Constant)

Sunspots are dark areas within the solar surface (photosphere) due to periodic disturbances and explosions in the solar surface. Number of sunspots **increases or decreases on a regular basis in a cycle of 11 years**. **When the sunspots appear in large numbers the intensity of solar radiation received at the surface is increased** (increase in solar constant). As the number of sunspots decreases, the quantity of radiation received at the Earth's surface declines (decrease in solar constant). But the percentage change in the solar constant associated with sunspots is rather negligible.

HEAT BUDGET

It is the **balance or equilibrium that exists between incoming solar radiation and outgoing terrestrial radiations**. The main source of energy for the Earth's climate system is incoming solar radiation. The sun emits energy in the form of shortwave radiation, which is absorbed by the Earth's atmosphere and surface. This absorbed energy is then re-emitted as long wave radiation (as infrared radiations). The Earth as a whole does not accumulate or lose heat and it **maintains its temperature**. This can happen **only if the amount of heat received in the form of insolation equals the amount lost by the earth through terrestrial radiation.**

Components of Heat Budget

1) Insolation

Insolation refers to the **incoming shortwave solar radiation** to the earth's surface. The processes involved with insolation in maintaining heat balance include:

- **Reflection** - Reflection occurs when incoming solar waves **bounce back from a surface** that it strikes in the atmosphere, on land, or water, and are not transformed into heat.
- **Absorption** - Absorption of radiation involves the conversion of electromagnetic radiation into **heat energy**.

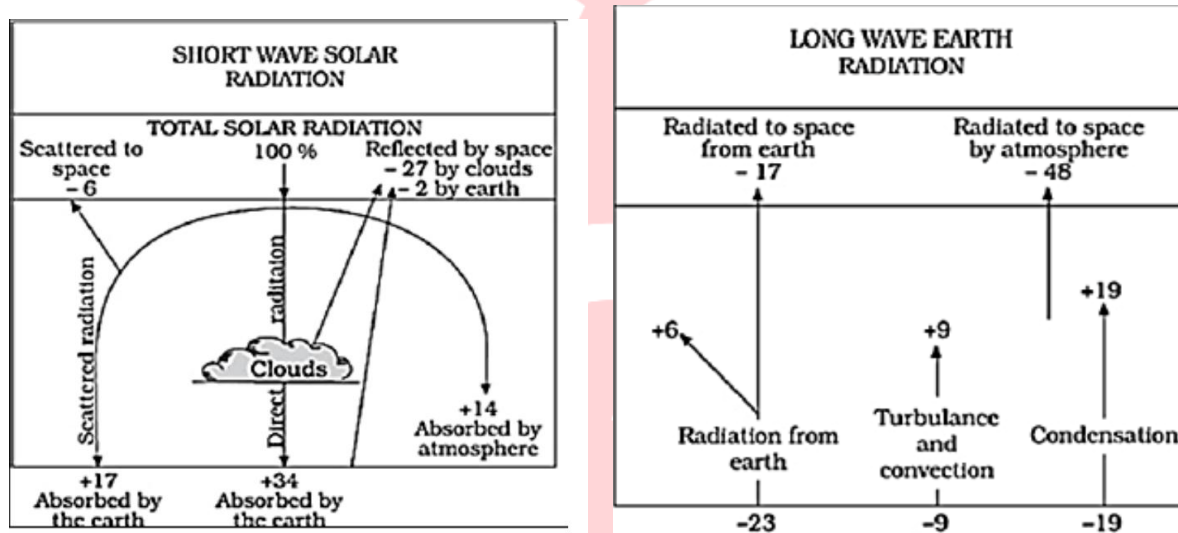
- **Scattering** - Scattering of solar waves takes place when the **radiation strikes small objects in Earth's atmosphere**, such as air molecules or water droplets or aerosols which disperse the solar waves in all directions.

2) Terrestrial Radiation

Terrestrial Radiation refers to **long wave radiation** that is emitted by the Earth's surface or by the atmosphere. The processes involved with Terrestrial Radiation in maintaining heat balance include:

- **Emission by radiation.**
- **Sensible heat transfer (conduction and convection)** - It is the energy that is transferred as heat to an object, without any change in the state.
- **Latent heat transfer:** It is the amount of heat transferred when one substance is changing its state. Example: From gas to liquid.

INFLUX and OUTFLUX: Earth's Complete Heat Budget



If the total insolation received at the top of the atmosphere is considered to be **100 units**, the table below traces the complete journey - from entry to exit - as one continuous flow. Influx and outflux are shown together, not separately.

<u>What Happens</u>	<u>Units</u>	<u>Notes</u>
INFLUX : 100 units of solar radiation enter the atmosphere		
Lost to space before reaching Earth's surface		
Reflected by clouds	27 units	Bounced directly back to space from cloud tops
Reflected by ground surface (snow, ice, etc.)	2 units	Reflected from Earth's surface back to space
Scattered by minute atmospheric particles	6 units	Dust, aerosols disperse radiation back to space
Sub-total lost to space (Earth's Albedo)	35 units	27 + 2 + 6 = 35 units. These do not heat the Earth or atmosphere.

Absorbed by the atmosphere (from incoming solar radiation)		
Absorbed by ozone layer and water vapour in clouds	14 units	Directly heats the atmosphere from incoming solar radiation
Absorbed by Earth's surface		
Direct insolation at surface	34 units	Reaches Earth directly and is absorbed
Scattered insolation that still reaches surface	17 units	Scattered but eventually absorbed at surface
<u>Sub-total absorbed by Earth's surface</u>	<u>51 units</u>	34 + 17 = 51 units. Will be re-radiated as terrestrial radiation.
<u>INFLUX</u>	<u>35 (lost) + 14 (atmosphere) + 51 (surface) = 100</u>	
OUTFLUX Earth re-radiates 51 units as terrestrial (long-wave) radiation		
What goes directly from Earth's surface to space		
Long-wave radiation radiated directly to space	17 units	Escapes the atmosphere without being absorbed
What goes from Earth's surface into the atmosphere		
Approximately 23% of 51 units emitted as longwave radiation, of which 6 is captured by the atmosphere	6 units	Absorbed as longwave radiation by greenhouse gases
9% net transfer back to lower atmosphere by conduction and convection (turbulence)	9 units	Sensible heat transferred from surface to lower atmosphere
Remaining 19% returns through release of latent heat of condensation	19 units	Released when water vapour condenses to form clouds/rain
<u>Sub-total: atmosphere absorbs from Earth's surface</u>	6 + 9 + 19 = 34 units	
<u>OUTFLUX (surface)</u>	<u>17 (direct to space) + 34 (to atmosphere) = 51</u>	
ATMOSPHERE BALANCE Atmosphere radiates all it absorbed back to space		
Absorbed from incoming solar radiation (influx)	14 units	Absorbed during influx phase
Absorbed from terrestrial radiation (outflux)	34 units	Absorbed from Earth's surface radiation
<u>Total absorbed by atmosphere (both sources)</u>	<u>48 units</u>	14 + 34 = 48 units. All 48 are radiated back to space.

FINAL BALANCE		
From Earth's surface directly to space	17 units	Long-wave radiation bypassing atmosphere
From atmosphere to space	48 units	All radiation absorbed by atmosphere re-emitted to space
<u>TOTAL RETURNING TO SPACE</u>	<u>65 units</u>	<u>17 + 48 = 65 = 65 received from Sun.</u>

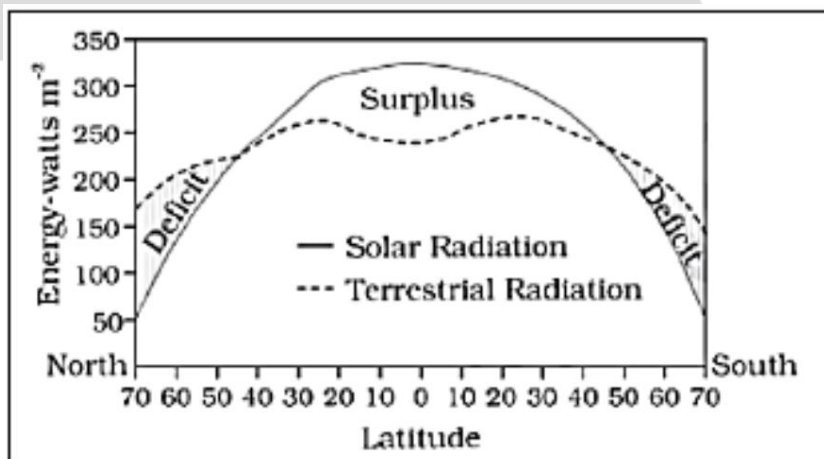
Out of the 51 units released from Earth in the form of terrestrial radiation, 17 units are radiated to space directly and the remaining **34 units are absorbed by the atmosphere** (6 + 9 + 19). The **48 units absorbed by the atmosphere** (14 units from insolation + 34 units from terrestrial radiation) are also radiated back into space. Thus, the total radiation returning from the earth and the atmosphere is **17 + 48 = 65 units which balance the total of 65 units received from the sun**. Therefore, the **Earth's heat energy budget is in balance**. This explains **why the earth neither warms up nor cools down despite the huge transfer of heat from the sun**. The **mean temperature of the earth is around 15°C at the surface**, although there are daily, seasonal and year to year changes. The **input in the system is the incoming shortwave solar radiation** that reaches Earth's surface; this is **balanced by the output of longwave terrestrial radiation** back to the atmosphere and to space.

HEAT BUDGET OF THE EARTH: IMPACT ON CLIMATE

The Earth's heat budget plays a critical role in determining the **planet's climate and weather patterns**. The balance between incoming solar radiation and outgoing long wave radiation, as well as the influence of cloud cover, surface temperature, and ocean currents, all contribute to the formation of global climate patterns. For example, the **presence of greenhouse gases in the atmosphere results in an increase in the Earth's average temperature**, leading to global warming and the potential for climatic disasters, such as sea level rise, tropical cyclones, and droughts. Understanding the Earth's heat budget and its impact on the planet's climate is essential in predicting and preparing for these events and in mitigating their effects.

LATITUDINAL HEAT BALANCE

The **difference between all incoming solar energy and all outgoing terrestrial energy by both shortwave and long wave radiation is called net radiation**. Even though the net radiation from the earth as a whole is theoretically zero, there are places where the solar energy received is more than the energy lost by terrestrial radiations (**energy surplus**



areas) and there are areas where the loss of energy through outgoing terrestrial radiation is faster than the gain of insolation (**energy deficit areas**). The **heat balance is for the overall earth atmosphere system and not for every latitude**. The incoming solar radiation at the surface of the Earth varies from place to place - some parts of the earth receive surplus radiation while some parts receive deficient radiation.

There is a **surplus of net radiation between 40°N and 40°S** and the **regions near the poles are in deficit**. The surplus heat energy from the tropics gets redistributed towards the poles in two ways: through **atmospheric circulations** (about 80% heat transfer) and through **ocean circulations** (about 20% heat transfer). This balance is extremely crucial as it ensures that the **tropics don't get extremely heated up** because of the accumulation of excess heat and the **areas in high latitudes do not get permanently frozen** because of deficit radiation.

FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE

Temperature refers to the degree of hotness or coldness. The major factors controlling the distribution of temperature include:

1. Latitude

The temperature of the atmosphere of a particular place near the ground surface mostly depends on the **amount of insolation** received at the place. The amount of insolation received by the ground surface **decreases from lower to higher latitudes** due to the decrease in the angle of incidence.

2. Altitude

Altitude refers to the **height above sea level**. Generally, **temperature decreases with increasing altitude** due to the **decrease in density of atmosphere**. The lower layer of air contains more water vapour than layers above and hence it absorbs larger amounts of heat radiated from the Earth's surface than the upper air layers. **Example:** The summit of Mount Everest, at an altitude of approximately **8,848 meters**, experiences much lower temperatures compared to the surrounding valleys due to its high elevation.

3. Distribution of Land and Water

Land heats and cools faster than water. There are three reasons for this phenomenon:

- The **specific heat of water is greater** than that of land.
- Water is transparent and solar energy can penetrate through the surface into the layers below, whereas **in opaque materials like soil and rock, solar energy is concentrated on the surface**. Thus, a given unit of heat energy is spread through a greater volume of water than land.
- Because **liquid water circulates and mixes**, it **can transfer heat to deeper layers**.

The **coastal areas** often experience a **narrower range of air temperatures** than locations in continental interiors, due to the **moderating influence of nearby oceans**. The coastal regions are moderated due to: the presence of **water in the vicinity** which warms and cools at a very slow pace; the presence of **land and sea breeze**, which helps in the mixing of air having different temperatures; and **ocean phenomena such as ocean currents and upwelling** which can also moderate the

coastal climate. **Example:** The city of Mumbai, located on the western coast of India, has a more moderate climate compared to the city of Nagpur, which is situated in the interior of the country, due to the moderating effect of the Arabian Sea.

The regions far away from the seacoast experience extremes of temperature, and the phenomenon is called continentality. Continental interiors are associated with higher annual range of temperature and daily range of temperatures. The difference between the highest and lowest temperatures of a year is called the annual range of temperature. The difference between the highest and lowest temperatures of a day is called diurnal or daily range of temperature.

4. Ocean Currents

Ocean currents can significantly impact the temperature of coastal regions. Warm currents, such as the Gulf Stream, can raise temperatures in adjacent land areas, while cold currents can lower temperatures:

- Alaskan current which is the extension of Kuroshio current, increases the temperature near the Northwestern coast of North America.
- Labrador current lowers the temperature near the North-Eastern coast of North America.

5. Winds

Winds help in redistribution of temperature. The winds blowing from lower latitudes to higher latitudes raise the temperature of regions where they reach while winds blowing from high latitudes to low latitudes lower the temperature. For example, westerly winds can bring warm air to temperate regions. Similarly, the sea breeze (sea to land) during daytime and land breeze (land to sea) at nighttime also helps in the redistribution of temperature in the coastal regions, which also helps in moderating the coastal climate. Local warm and cold winds can also alter the temperature of locations. For example, Loo, a local hot wind increases the temperature in the northern plains of India and can create heat wave situations. Cold local winds can decrease the temperature of a region.

6. Nature of Ground Slope

The ground slope facing the sun receives more insolation. In the northern hemisphere, the southward facing slopes of east-west stretching mountains receive greater amount of insolation than the northward facing slopes because of their exposure to the sun for longer duration. Similarly, the windward side of a mountain may receive more precipitation and have cooler temperatures, while the leeward side, or rain shadow, can be drier and warmer (will be explained in orographic precipitation).

7. Mountain Barriers

Mountain ranges block the movements of air from one place to another. For example, the Himalayas do not allow the cold winter air masses from Siberia to enter the Indian subcontinent, so that the winter temperatures of the entire region to the south of the mountains remain relatively higher than what they would have been in their absence.

8. Nature of the Surface

The nature of ground surface in terms of colour, vegetation and land use pattern such as urban infrastructure affects the distribution of temperature. Different kinds

of surface reflect different amounts of incoming solar radiation - **dark coloured surfaces absorb more insolation than the light-coloured ones**. The snow-covered surfaces absorb very low amount of insolation because they reflect **70 to 90%** of the insolation. Areas with **dense vegetation tend to have cooler temperatures** due to evapotranspiration, while **urban areas often experience the urban heat island effect**, leading to higher temperatures (to be discussed in future).

Previous Year Question

1. Consider the following ecosystems: [2002]
1. Taiga 2. Tropical evergreen 3. Tropical deciduous 4. Tundra
The correct sequence in decreasing order of the albedo values of these ecosystems is -
a) 1, 4, 3, 2 b) 4, 1, 2, 3 c) 4, 1, 3, 2 d) 1, 4, 2, 3
2. Assertion (A): The thickness of the atmosphere is maximum over the Equator.
Reason (R): High insolation and strong convection currents occur over the Equator. [2004]
a) Both A and R are individually true and R is the correct explanation of A
b) Both A and R are individually true but R is not the correct explanation of A
c) A is true but R is false
d) A is false but R is true
3. Consider the following statements: [2008]
1. The albedo of an object determines its visual brightness when viewed with reflected light.
2. The albedo of Mercury is much greater than the albedo of the Earth.
Which of the statements given above is/are correct?
a) 1 Only b) 2 only c) Both 1 and 2 d) Neither 1 Nor 2
4. Which one of the following reflects back more sunlight as compared to the other three? [2010]
a) Sand desert b) Paddy crop land c) Land covered with fresh snow d) Prairie land
5. Consider the following statements: [2022]
1. High clouds primarily reflect solar radiation and cool the surface of the Earth.
2. Low clouds have a high absorption of infrared radiation emanating from the Earth's surface and thus cause a warming effect.
Which of the statements given above is/are correct?
a) 1 only b) 2 only c) Both 1 and 2 d) Neither 1 nor 2
6. Consider the following statements: [2024]
Statement-I: The atmosphere is heated more by incoming solar radiation than by terrestrial radiation.
Statement-II: Carbon dioxide and other greenhouse gases in the atmosphere are good absorbers of long wave radiation.
Which one of the following is correct in respect of the above statements?
a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I

- c) Statement-I is correct, but Statement-II is incorrect
- d) Statement-I is incorrect, but Statement-II is correct

7. Consider the following statements: [2024]

Statement-I: Thickness of the troposphere at the equator is much greater as compared to poles.

Statement-II: At the equator, heat is transported to great heights by strong convectional currents.

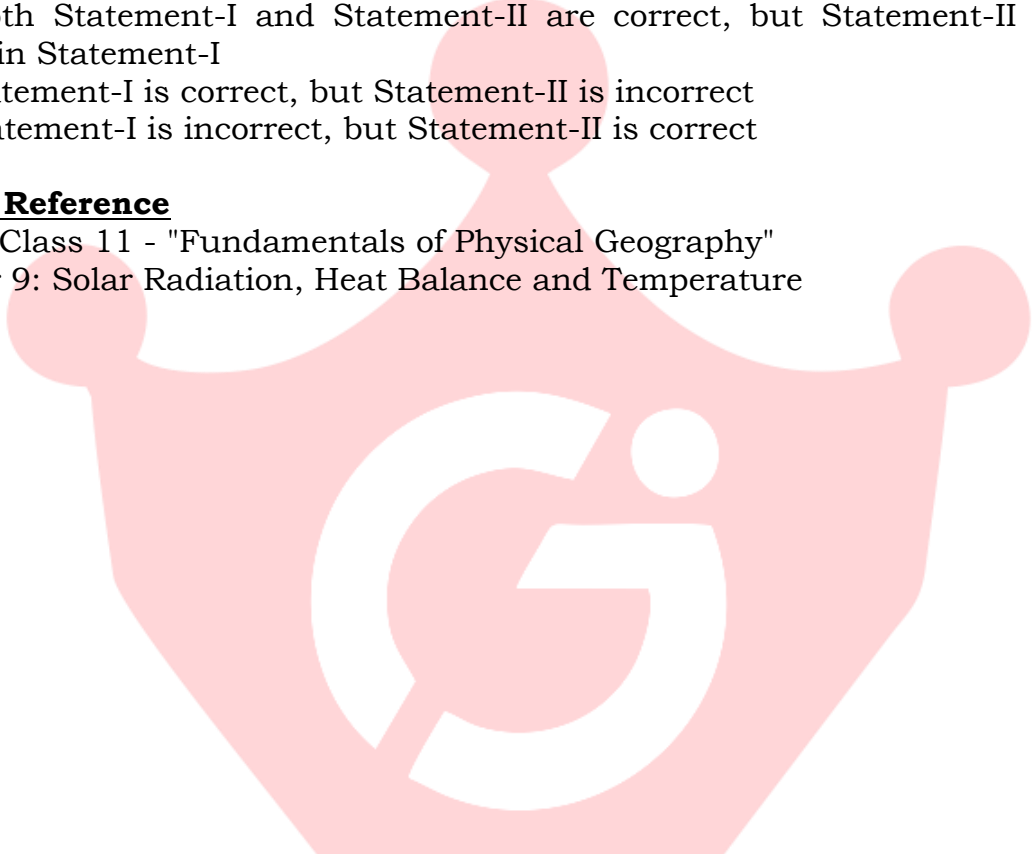
Which one of the following is correct in respect of the above statements?

- a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
- b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
- c) Statement-I is correct, but Statement-II is incorrect
- d) Statement-I is incorrect, but Statement-II is correct

NCERT Reference

NCERT Class 11 - "Fundamentals of Physical Geography"

Chapter 9: Solar Radiation, Heat Balance and Temperature



GALLANT IAS